THE UNIVERSE OF DRONES

Hunting in packs and crippling air defences, these are some of the world's most lethal drones



IRAN

SHAHED-136

The Iranian drone that has spread alarm in adversaries' ranks and spawned many imitations

- **COMBAT RADIUS:** 1,000–2,500 km; endurance of 6 hours
- **WARHEAD:** 30–50 kg
- **TACTICAL USE:** Mass swarm capability, it has had proven impact in the Ukraine war. Known as the "poor man's cruise missile"

CHINA

SUNFLOWER 200

Chinese-made delta-wing, low-cost kamikaze drone usable in swarms; considered to be an upgrade of the Iranian Shahed, with similar capabilities

- **COMBAT RADIUS:** 1,500-2,000 km; endurance of 10 hrs
- **WARHEAD:** 40 kg
- **TACTICAL USE:** Designed to hit long-range stationary targets

ISRAEL

IAI HARPY



Israeli kamikaze anti-radiation loitering munition; can be used in swarms

- **COMBAT RADIUS:** 500 km,endurance of 9 hrs
- **WARHEAD:** 30+ kg
- **TACTICAL USE:** Detects, tracks and destroys enemy radar systems



UKRAINE

AQ-400 SCYTHE

Long-range strike UAV developed for deep missions in contested environments by Ukraine during its war against Russia. Used in swarms, it has proven its worth in operational scenarios

- **COMBAT RADIUS:** 750 km; endurance of 7 hrs
- **WARHEAD:** 43 kg
- **TACTICAL USE:** Deep strike into rear bases, logistics hubs

future warfare. Today, there is a need to create such a drone manufacturing ecosystem in India that is completely self-reliant—not only at the product level—but also at the component level so that the drone's mould, software, engine and batteries are all made in India." He promised manufacturers "every kind of support from the government".

"Air warfare is shifting from piloted systems to unmanned platforms, and we must accelerate our efforts," says Lt Gen. Raj Shukla (retd), former commander of Army Training Command. "We have not been able to keep pace, though we have enough talent and capability in the drone manufacturing ecosystem."

Operation Sindoor—the 87-hour aerial clash with Pakistan in May 2025—delivered a brutal wake-up call to India's drone forces. Pakistani directional jamming arrays—where high-power radio frequency waves disrupt GPS-reliant communication/ navigation—and other EW tactics like signal spoofing crippled much of India's UAV inventory. Many Indian platforms, reliant on vulnerable radio frequencies and GPS links, could not operate effectively.

Indeed, before Operation Sindoor, EW received scant attention in Indian drone procurement. A reliance on imported subsystems—sensors, communication kits, propulsion and precision electronics—left platforms exposed to signal denial and jamming, turning them into deadweight duds. Clearly, combating EW is a major challenge, for without protection against it, even advanced drones can fail in combat.

Countering the EW Threat

In electronic warfare, jamming usually targets control and video. Disrupting these is sufficient to cut off operator control and eliminate real-time situational awareness. GPS-jamming is a more aggressive step. It can degrade navigation, trigger fail-safe modes (return-to-home) or reduce positional accuracy. During Op. Sindoor, Indian drones could not counter the basic disruption of control and video links, revealing limited anti-jamming capability, fragile communication links and insufficient autonomy when connectivity is severed.

To counter EW, modern drones combine command, telemetry and video into unified digital links rather than through separate channels. Advanced models also employ frequency hopping spread-spectrum (FHSS) techniques—which rapidly change frequencies—and encryption to resist basic jamming. Some can maintain autonomous flight or switch to other backup behaviours when links are lost.

An expert on drones points out that, with India finally adopting coordinated swarms of drones as its drone strategy, it must now develop distributed drone intelligence and algorithms that ensure seamless operation even under disruption. Such decentralised algorithms